FAST REACTOR DEVELOPMENT IN THE UNITED STATES

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THE advanced fast reactor programme in the United States consists of parallel efforts on several concepts. The concepts receiving the most attention are the sodium cooled fast ceramic reactor, the gas cooled fast reactor, and the molten plutonium reactor.

1. The Fast Ceramic Reactor Programme

Fuelling a large fast power reactor with mixed oxide (PuO_2-UO_2) provides the potential for conveniently combining the high irradiation performance characteristics of ceramic fuel with the inherent safety due to the Doppler effect in a relatively dilute, softened spectrum reactor system. Demonstration of this potential for low power cost and safe, reliable operation has been the continuing objective of the three major parts of the fast ceramic reactor programme. These are:

(1) the FCR development programme, devoted to developing the requisite fuel technology, reactor physics, and core engineering needed for integration into a feasible reactor concept;

(2) the SEFOR project, an experimental reactor test of FCR safety and dynamics characteristics;[1] and

(3) the sodium mass transfer programme, a complementary investigation of corrosion and mass transport in sodium coolant systems being conducted as part of the sodium component development programme.[2]

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Fuel Performance and Design

The capability of mixed oxide fuel to achieve high burnup (up to 100,000 MWd/t) at high power rating (up to 23 kW/ft) was shown with both sintered pellet and swaged powder fuel specimens early in the FCR development programme.[3] The high burnup contributes to attractively low fabrication, processing and out-of-reactor inventory costs; the high power rating, to low in-reactor inventory and first core capitalization costs.

The fuel also exhibits useful resistance to short-term operation under off-standard conditions arising either from the fuel itself or from the reactor. Specifically, intentionally defected fuel specimens have been shown to withstand thermal cycling following sodium logging without further clad failure or deformation.[4] Similar resistance has been shown by zero burnup fuel to transient overpower conditions which produced gross fuel melting.[5] Preliminary results with pre-irradiated fuel indicates a possible fission gas pressure stress problem if fuel melting during the transient is permitted beyond the region from which gas has been released during steady state operation.[6]

While the thermal flux irradiation programme continues with the testing of mixed oxide fuel under increasingly rigorous conditions, it is now being supplemented by fast flux irradiations of both fuel and clad specimens in EBR-II. Such fast flux irradiations more closely simulate the FCR conditions with respect to flux, temperature topography, and fission product spectrum, in the case of fuel specimens; fast neutron damage and induced H and He effects, in the case of clad materials. Initial, short-term ($<2\times10^3$ MWd/t) fast flux irradiations[7] yielded results comparable to those achieved in thermal flux irradiations, and served primarily to check flux and thermal conditions used in the design of fuel pin and clad material capsules now in EBR-II for scheduled long-term (50–100×10^3 MWd/t) irradiation. Additional irradiations in both EBR-II and the Fermi reactor are planned.

Venting of fission gas (Kr and Xe), a large fraction of which is released from high burnup–high power density FCR fuel, is required to prevent excessive pressure build-up within a fuel pin. This can be accomplished by venting either internally, for example to an end plenum, or externally to the coolant. The vented-to-plenum design was initially favoured despite its large plenum–pressurized clad requirements because of the concern that long-lived fission product contamination might escape along with the gas on venting-to-coolant. However, in a preliminary test of a vented-to-coolant fuel pin of simple, diving bell design,[8] the release fractions of all non-gaseous nuclides (including I-131, Cs-137, and Pu) were found to be at or near the limit of detection (10^{-6} to 10^{-10}). Further, although the release of fission gas was at the predicted level, the hold-up time (approx. 5 days) was such that Cs-134, an activation product of the 5·3 d Xe-133-stable Cs-133 chain, was the only activity found in significant quantity in the surrounding sodium, after Na-24 decay. In addition, no deleterious effects to the fuel were observed due to possible exposure (16×10^3 MWd/t) to sodium vapour.

On the basis of the foregoing results and a comparative evaluation of safety factors,[9] the vented-to-coolant fuel design is now preferred as the reference choice. By relieving the pressure stress on the fuel clad, a reduction in failure frequency on irradiation to high burnup is expected. Although quantitative prediction of the improvement is not possible, the resulting reduction in coolant contamination from the leaching of failed fuel may well more than offset the slight contribution from venting. Also, the threat of fuel pin failure propagation within a subassembly should be reduced by low internal pressure. The possibility that fission gas under high pressure could be ejected into the sodium coolant stream through a minor defect, causing vapour blanketing and clad failure of adjacent pins and perhaps choking off of coolant flow, is effectively eliminated. Such propagation would be objectionable not only relative to fuel failure itself, but also could produce a serious reactivity effect due to sodium voiding if sufficiently widespread.

Core Engineering and Reactor Design

Periodically during the FCR development programme, a large reactor has been conceptually designed, incorporating the then current status of fast reactor physics, core analysis, and fuel and coolant technology. From an initial 300 MW(e) design,[10] these studies have escalated in scale to the present 1000 MW(e).[11, 12, 13]

From the outset, attention in these studies has been focused on satisfying conservative safety criteria while capitalizing on the high performance capability of mixed oxide fuel and sodium coolant. In a large fast reactor, the fuel Doppler effect provides the major source of prompt negative reactivity coefficient, and thus a large Doppler value is essential to assure stability and excursion limiting capability. The low enrichment characteristics of a large reactor core and the softening of the fast neutron spectrum by the oxygen in the FCR fuel both tend to increase the Doppler coefficient. To further degrade the spectrum and thus enhance the Doppler effect, BeO can be deliberately added to the core at some expense to neutron economy. In the latest 1000 MW(e) FCR design,[13] variable BeO addition in two radial core zones is proposed: 10·5 vol.% in the central zone, 3·6 vol.% in the outer zone. The resulting power flattening was further improved by a BeO radial reflector surrounding the core. A substantial (0·006) negative Doppler effect ($T\{dk/dT\}$) was calculated even with total sodium loss for this design.

Accidental sodium voiding from the centre of a large fast reactor could constitute a serious safety problem, threatening as it does a positive reactivity insertion of substantial magnitude and rate. Attacks on this problem have largely involved high leakage core configurations, i.e. annular, modular or low height to diameter ratio cores.[14] The low H/D or pancaked core has been selected for FCR designs because of its effectiveness in achieving favourable dynamics with comparative physical and mechanical simplicity. In the latest 1000MW(e) FCR design study, a severely pancaked core (16 in. height × 11·5 ft diameter, with 18 in. thick top and bottom axial blankets) was required to satisfy the sodium loss criterion: total loss from core and axial blankets to be zero or negative.

Despite the concessions made in core composition and configuration to satisfy minimum Doppler and maximum sodium void safety criteria, the resulting design had a breeding ratio of 1·18 and a specific power of 1010 kW/kg fissile Pu + U, giving a doubling time of 13·4 years. A fuel cycle cost of 0·9 mills/kW h was calculated using a conservative set of nuclear data and conventional fuel cycle costing assumptions. Coupled with previously calculated[12] capital ($125/kW) and operating costs of 2·5 and 0·3 mills/kWh respectively, a total power cost of 3·7 mills/kW h is projected for this 1000 MW(e) FCR design.

SEFOR Project

The south-west experimental fast oxide reactor (SEFOR) programme is intended to verify experimentally the calculated behaviour of the Doppler effect and related dynamic characteristics basic to the stable and safe operation of large FCR's.[1, 15] Approximately half of the $25 million cost of this project will be incurred in reactor design and construction, provided by the Southwest Atomic Energy Associates (a group of seventeen investor-owned electrical utilities located in the south-western United States), the Gesellschaft für Kernforschung mbH of West Germany, Euratom, and the General Electric Company. The balance of the cost is provided by the U.S. Atomic Energy Commission to provide for necessary research and development.

Construction of this 20 MW(th) test reactor is now in progress near Fayetteville, Arkansas, with completion scheduled for late 1957 preparatory to a planned 3-year operating period. SEFOR is designed to have physics characteristics similar to those calculated for large FCR's and will operate with fuel composition and temperatures comparable to such reactors. Experiments are planned for SEFOR[16] to obtain power coefficient data over a broad range of power conditions: from steady state, through low and high frequency oscillations, to fast transient tests, both subprompt and prompt critical.

The engineering and associated research and development activities are proceeding satisfactorily with no technical obstacles apparent to date which jeopardize attainment of the programme goals. Fuel has been successfully tested in GETR and TREAT over the expected range of steady state and transient operating conditions; the first in a series of critical experiments has been completed in ZPR-III; and tests of critical components (instrumented fuel assemblies, fast reactivity excursion device, refuelling cell, etc.) are being conducted.

Future Prospects

The fuel performance and reactor safety design data being obtained should provide a basis for a decision to commit a prototypical 300 MW(e) FCR. In the interim, it will be necessary to achieve continued successful implementation of the FCR development programme, including fast flux irradiations, over 1 year's satisfactory operation of SEFOR, and a parallel plant design effort to establish the design characteristics, total power generation costs, and additionally required applied development for the 300 MW(e) FCR. Experience gained with this prototype FCR, and similar reactors in the United Kingdom and the U.S.S.R., could in turn provide the basis for design of a large plant in the 1000 MW(e) range for construction in the latter half of the 1970's. If these evolutionary steps confirm the promising economic and safety features calculated for the FCR, this reactor type could well provide the means for deep penetration by fast breeder reactors into the total electric power generating capacity by the end of this century.

2. Gas Cooled Fast Reactors

Background of the Project

Current USAEC interest in gas cooling of fast reactors stems from review of a feasibility study that was conducted during 1962 and 1963 at General Atomic Europe in Zurich, Switzerland.

In 1963 this was continued by USAEC sponsored work, concentrating on core design studies in the 400 MW(e)–1000 MW(e) range. These studies were aimed at assessing the practicability of the gas cooled fast reactor (GCFR) concept and included investigation of core performance as a function of size, analysis of potentially significant accident conditions associated with the possibility of steam entry or loss of coolant, determination of criteria for fuel element design, an initial estimate of fuel cycle economics, and an assessment of the necessary research and development programme to demonstrate the concept.

Subsequent AEC sponsored work was concerned with conceptual design studies of a reactor experiment to develop the GCFR concept, with the initiation of fuel element development, and with investigation of the future potential of the system.

Current work is aimed principally at initial experimentation associated with fuel element development and with a more detailed safety analysis of the system. Concurrent with USAEC activities, a privately funded study is directed towards the design and cost evaluation of large, approximately 1000 MW(e), gas cooled fast reactors. This study team is composed of General Atomic associated with fourteen different utility companies of the East Central Nuclear Group.

Although this paper is limited to a brief summary of GCFR development, past work is described in greater detail in refs. (17)–(20).

Basis of Interest

The initial interest in the development of gas cooled fast breeder reactors arose in part from the early determined fact that all the cooling needed could be obtained with modest pumping powers and reasonable gas pressures (~ 1000 lb/in^2), and in part from the inherent passivity of gas as a fast reactor coolant—namely, with regard to the chemical inertness of helium and the minimal neutron scattering and spectrum degradation obtained with a low density coolant.

For example, the worry of severe reactivity insertions from coolant loss is eliminated due to the low reactivity associated with the coolant and, moreover, due to the impossibility of sudden void formation in a gas. However, studies have shown that the coolant coefficient, although small, can be important in controlling transients. A reference design by ORNL[21] of a 2500 MW(e) (Pu, U)O_2 helium cooled reactor has a reactivity gain of 0·0051 δk (\$1·47) upon complete loss of coolant. The Doppler coefficient of $-0{\cdot}4 \times 10^{-5}$ $\delta k/°C$ provides protection such that core melting does not occur if three of the four blowers stop and coolant pressure is simultaneously reduced to atmospheric in 20 sec.

The neutronic passivity of the helium coolant makes it possible to utilize low leakage cores with high internal conversion ratios. As a result, batch fuel management is specified for GCFR's, with fuel lifetimes of up to 3 years between complete core reloadings, which nevertheless requires less than one dollar reactivity swing during lifetime.

It is clear that the harder spectrum with gas cooling and therefore higher breeding ratio gives the possibility of reduced fuel cycle costs. Current core design and safety studies, moreover, indicate that no severe compromises are needed which would degrade the economic potential (i.e. extra moderation for reasons of safety).

Reference Design Configuration and Principal Components

Details on the mechanical aspects of this design are discussed in ref. (18). Very briefly, the system is characterized by the following features:

(1) A 5000 litre core with 0·5 length to diameter ratio surrounded by a 46 cm thick reflector.
(2) Containment of the entire primary circuit in a prestressed concrete pressure vessel.
(3) Axial-flow helium circulators driven by single-stage steam turbines.
(4) Once-through steam generators in multiple parallel arrangement.
(5) Cantilever suspension of fuel assemblies in a 4 ft deep grid plate with downward coolant flow.
(6) $B_4^{10}C$ control rods operating from the top provided with pneumatic scram.
(7) Normal fuel replacement from the bottom in dry conditions.
(8) Provision for emergency cooling and fuel replacement under water flooding conditions with reactivity control by incorporation of thermal and resonance poisons.
(9) Use of partially roughened surfaces on the fuel cans to aid heat transfer.
(10) Core arrangement to minimize voidage (square pitch, bottom loading, etc.).

Development Programme

Principal steps of the development programme proposed for this concept are: first, construction of a helium cooled reactor experiment in the 150 MW(th) range; second, construction of a prototype in the 250–500 MW(e) range; and third, construction of a demonstration plant in the 1000 MW(e) range.

The experimental implementation of the development programme has begun with General Atomic–Oak Ridge collaboration on fuel capsule irradiations. Tests of clad oxide fuel are under way in the ORR poolside facility, and other tests in ORR gas cooled loop No. 1 are planned. These irradiations are being conducted in a predominantly thermal–neutron spectrum and are designed primarily to investigate fuel cladding interactions.

The two fuel element concepts being investigated consist of cylindrical Hastelloy-X or stainless steel tubes, approximately 1 cm in diameter, containing UO_2 pellets. In one concept the clad is fuel-supported and is designed to collapse on to the fuel pellets with time after being subjected to the test temperature and pressure. The pellets are specially shaped to reduce the effects of fuel cladding interaction. The other element is designed to minimize the effect of system pressure. One version of this concept has a specially deformed unfuelled upper region, or flex-tube that is capable of flexing as a bellows to transmit the reactor coolant system pressure to the inside of the element, and thereby it may permit the use of free-standing cladding in the fuel region.

The current exploratory thermal-flux irradiations are of short duration ($<10{,}000$ MWd/t) and are being carried out at a linear heat rating of up to 20 kW/ft, 1400°F (760°C) cladding temperature, and 800–1000 lb/in^2 external pressure (loop 1 tests will be at lower pressure). The poolside capsules are subjected to thermal cycles by moving the capsule with respect to the reactor. Results are revealing behaviour mechanisms and indicating improvements that can be made in design features. When present designs are sufficiently refined, tests of longer duration (up to 100,000 MWd/t) will be conducted in fast flux test facilities such as EBR-II and Fermi.

Reactor Experiment

The conceptual design of a reactor experiment was completed as part of the GCFR work this past year. This experiment was designed as a combined physics, dynamics, and fuel element test facility.

For this experimental reactor it is proposed to use oxide fuel with the same ratio of uranium to plutonium as that of the prototype power reactor in order to match neutronic characteristics, such as the neutron balance, the spectrum, and the Doppler coefficient.

Fuel pins are of the same dimension as those of the prototype in order to match heat transfer characteristics. The core size, however, has been reduced to approximately 2000 litres rather than the 5000 litres of the reference design by reduction in coolant volume fraction and power density.

The result is a system that produces 150 MW(th), or roughly one-quarter of the reference design heat rating for the fuel pins. The operation of fuel at this lower rating puts it in the realm of conventional practice and is the principal design factor making it possible to suggest construction of a reactor experiment at an early date.

Initial experiments would be reactor physics investigations and determination of the dynamic behaviour of the core for a range of power levels.

Following this phase, the reactor experiment will be utilized as a fuel test facility for GCFR elements by incorporating a central zone in which more highly enriched fuel elements would operate at full prototype power density. Other core loadings are possible which would result in high fluxes for accelerated irradiation experiments.

Future Potential

The bulk of preliminary studies of the prospects of the gas cooled fast reactor system were confined to what could be done with helium as the coolant, at 1000 lb/in^2 working pressure, with oxide fuel contained in cans limited to a design surface temperature maximum (before hot spot allowance) of 1220°F (600°C). Broadly, the conclusions were that with gas temperatures and pumping expenditures set to allow overall plant efficiencies of 38–40%, fuel ratings of the order of $\frac{3}{4}$ MW(th)/kg fissile were possible, together with a conversion ratio of about 1·5, yielding system doubling times of 10 years or so, and fuel cycle costs on the order of 0·7 mills/kWh(e) under private fuel ownership or 0·3 mills/kWh(e) under public ownership.

The extremely low fuel cycle cost indicated for advanced GCFR designs also suggests application where large quantities of heat are required such as evaporative desalination plants. Moreover, initial investigations of single-purpose plants[19] show also that reactor performance is considerably enhanced by the lower gas inlet temperatures required.

Starting primarily from an interest in seeing whether liquid metals were essential to fast reactor cooling, it was found that the cooling needed can be obtained economically with gas and, moreover, that there are a number of direct and side benefits of gas cooling—particularly in large systems.

These favourable indications are based on studies of fuel materials selected for present familiarity with them rather than for their ultimate potential, and, in line with this outlook, plant design has centred around maximum use of equipment development for other reactor types—particularly the HTGR system.

Although the idea of a gas cooled reactor initially seemed a novelty, it now appears, based on initial studies, technically feasible and economically attractive.

3. Los Alamos Molten Plutonium Programme

Molten plutonium alloys are among the advanced fast reactor fuels being investigated for possible application to central-station power generation.[22] There are major potential advantages for molten plutonium fuels in the areas of safety and stability, fuel element lifetime, and doubling time.

It was recognized early in the development of molten plutonium fuel that containment would be difficult. In principle, the use of a fluid fuel permits the changing of fuel without the replacement of the container. The design decision as to whether to refill or to replace containers, however, must be based on consideration of the lifetime of the container as compared with that of the fuel. For a system in which molten plutonium is contained in tantalum, it is not clear whether the fuel or the container lifetime (if either) will ultimately be limiting.

The designs for molten plutonium experiments now planned use sealed capsules but are compatible with the use of vented capsules. The selection of a capsule-type fuel containment has several distinct practical advantages: the technology developed for the thermal and mechanical design of solid fuel elements is available and useful; a failure of containment has limited and tolerable consequences; and the design is well suited to collection of data on properties of materials as a function of burnup. One disadvantage is that the container is required to be a heat transfer medium.

The basic feasibility of using molten plutonium as a fast reactor fuel is illustrated by the more than 2 years of successful operation of the 1 MW LAMPRE I reactor.[23] This operation demonstrated:[24] (1) the capability of operating with a molten fuel, in this case the Pu–Fe eutectic alloy in sealed capsules; (2) the release of a large part (80%) of the volatile fission products from the fuel to the contained gas space; (3) the nuclear stability of this type of system; and (4) the containment of the fuel for extended periods of time.

Current emphasis in the molten plutonium programme is on questions relating to properties of fuel (and containment) during and after high burnup. Pertinent data will be provided by a continuing series of irradiation experiments presently under way in the thermal flux of the Los Alamos Omega West Reactor (OWR),[25] as well as by extensive out-of-pile tests.

Present Status

(a) Fuel

Pu–Fe fuel contained in tantalum capsules, as in LAMPRE I, is not well suited to a large central-station fast breeder application, because of its high plutonium density and consequent low attainable specific power. For a core in which the container is simultaneously a heat transfer medium and the dominant parasite, it is necessary to compromise between high specific power (requiring a large ratio of heat transfer area to fuel mass) and good breeding potential (requiring a small ratio of parasitic mass to fuel mass).

The quantitative implications of this compromise depend, of course, on the materials and dimensions involved. For a core, composed of tantalum capsules of reasonable dimension, fuel containing from 3 to 6 g Pu/cm^3 of fuel appears to give a minimum doubling time.[26] If suitable containment can be achieved with a lower parasitic loss rate, the optimum design would utilize a lower plutonium density in the fuel. The relationship between container parasitic cross-section and optimum plutonium density is significant in that the lower plutonium densities appear to be easier to contain. Thus, there may be potential for performance improvement in this direction.

Present development effort is directed towards ternary alloys containing Pu, Co, and Ce. These alloys offer two major advantages over the Pu–Fe of LAMPRE I: (1) the plutonium concentration is variable (over the entire range of alloys from 0 to 13 g Pu/cm^3 fuel, the melting point is less than 842°F [450°C]), and (2) the fuel is far less corrosive to the selected containment materials than is Pu–Fe at equivalent test conditions.

Coupled with the advantages of these molten alloys, three areas are of immediate concern. The first, and most obvious, is the integrity of the container. Present experience indicates that tantalum or Ta–W alloys are the most suitable container materials for these fuels. The development effort which has been expended on this facet has resulted in improvements to the extent that reliable long term fuel containment may be anticipated. The second problem area with Pu–Co–Ce fuels is that they exhibit large volumetric expansions during freezing (1–3% at plutonium concentrations from 0 to 8 g Pu/cm^3). This characteristic poses a potential operational inconvenience in that the core may have to be maintained in a molten condition at all times or frozen only under controlled conditions, if mechanical damage to the container is to be avoided. The third area is associated with the formation of an intermetallic reaction layer between a component of the fuel (Co) and the container (Ta). If this layer is damaged during irradiation, undesirable mass transport effects may occur. Experiments are under way to obtain data on this effect.

(*b*) *Containment*

The dominant corrosion-failure mechanism involved with liquid plutonium alloys contained in refractory metals is that of intergranular penetration of the plutonium through the grain boundaries of the container.[27] Containment lifetime, as measured by length of time before a fuel element shows detectable plutonium on the surface, can vary for apparently equivalent test conditions; the ratio of longest observed lifetime to shortest in early tests was greater than 10.

Since there have been few failures with the Pu–Co–Ce fuel, no definite conclusion can be drawn concerning expected lifetime of this fuel system. However, there is substantial evidence that significant corrosion of tantalum or Ta–W alloys will not occur at temperatures of 1562°F (850°C) or lower, except perhaps in welds. It has been found that a carburized layer on the internal surfaces of the container, or the diffusion of carbon into the tantalum grain boundaries, is effective in inhibiting plutonium penetration with the more corrosive Pu–Fe fuel.[28] Evidence also indicates the latter treatment to be effective in inhibiting penetration of welds by Pu–Co–Ce fuel.

Tungsten additions to tantalum have pronounced effects on hardness, ultimate and yield strengths, and total per cent elongation. Since container strength is important for preventing capsule distortion in the event of inadvertent freezing of the fuel, and since tungsten additions to tantalum may be beneficial to corrosion performance, the present reference container material is Ta-5W.

(*c*) *Irradiation effects*

With effective containment in the absence of radiation apparently developed for the Pu–Co–Ce fuel system, we may turn to the next major concern: the behaviour of the fuel and of the containment after long irradiation. Even the mechanism which will limit burnup capability is not apparent at this time. No hint of any serious irradiation limit (except the build-up of fission gas pressure in the sealed capsules) was apparent from LAMPRE I.

Initial thermal flux irradiation experiments are under way in the OWR. The current OWR experiment series was designed to investigate wall effects in tantalum which may arise from fission product recoils. The average burnup in the first irradiation was approximately $2{\cdot}5 \times 10^{19}$ fissions/cm^3 or 0·15 at. % Pu burnup. The fission density at the wall is calculated to be six times the average. The second capsule, identical to the first, was irradiated to approximately 1 at. % Pu average burnup, or $1{\cdot}6 \times 10^{20}$ fissions/cm^3 fuel. Again, the fission density at the wall was six times the average. In neither of these tests was significant wall damage observed. An additional observation is that in these two capsules, which were loaded with the same amount of fuel, the final fuel heights were the same (to within a small fraction of a millimetre) after irradiation. This result suggests that fission gas release reached a steady state within the 2-week operational period of the first capsule and that the volume per cent of bubbles within the fuel remained constant after initial transient conditions.

The next experiments in the OWR will compare the irradiation effect on Ta-5W containers with and without an internal carbide coating. It is planned that these capsules will be run to approximately 3% average Pu burnup.

(d) Large core

A concept for utilizing molten plutonium in a large central-station fast breeder was discussed in ref. (29). The general configuration assumed is an array of reactor modules, containing a sufficient number of modules to produce 500–1500 MW(e) (perhaps 10–30 modules). A specific and realistic description of a molten plutonium fuelled capsule core, using current best estimates for limiting factors, yields a doubling time of 7 years, breeding ratio of 1·5, and specific power of 1 MW/kg Pu. This breeding ratio is perhaps surprising, considering the high tantalum inventory. The favourable breeding potential is largely due to the low parasitic capture in equilibrium plutonium which is associated with the hard neutron spectrum. For each 100 neutrons causing fission in plutonium in the molten plutonium fuel, approximately 10 are parasitically captured in plutonium, 25 in tantalum. For a typical ceramic core, comparable numbers are 35 parasitic plutonium captures and 15 non-fuel parasitic losses.[30] It is also clear that a large margin for improvement of the molten plutonium systems is available, if a suitable container with a lower parasitic cross-section can be found.

Safety

The dominant inherent shutdown mechanism in a molten plutonium fuelled core is the large thermal expansion coefficient of the fuel. Calculations for

LAMPRE I indicate that with non-irradiated fuel, a reactivity ramp of \$100/sec (20% Δk/sec) could be tolerated without damage to the tantalum containers. For extreme transients, entrained fission product gas bubbles (which will be present in any irradiated molten fuel) increase the inertial delays in the expansion of the fuel. For the same reactivity input, the energy integral and temperature rises would then be greater than if no bubbles had been present. On the other hand, the cushioning effect of the bubbles is such that, even with the greater energy integral, the pressure rise is calculated to be less than it would be if no bubbles were present.[(29)] Since it appears that transient over-temperatures are easily handled by these systems, the presence of bubbles will be a safety advantage. A similar range of safe response is expected for large cores using molten plutonium fuel. Consideration is being given to the use of a fast burst facility for generating transients suitable for the investigation of those areas which might be limiting for molten plutonium capsules.

Several further safety advantages are present for a molten plutonium fuelled fast breeder. The use of tantalum containment requires that the system utilize an extremely hard neutron spectrum, to avoid excessive parasitic loss.[(32)] This, in turn, leads to small, totally external breeder designs, with a large power system being obtained by a coupled (e.g. modular) arrangement.[(29)] As is well known, these systems have strongly negative sodium void effects. Further, since the fuel is normally operated in a molten state, there is no concern for the fuel slump which can accompany fuel melting in a solid fuel core. The tantalum temperatures during normal operation are far below those which would cause short term damage, so that transient overheating of the core is not of concern. For example, \$50 or more of reactivity would have to be inserted to raise the temperature of the core to the melting point of tantalum (5340°F or ~3000°C). Since these are the factors which are commonly considered as contributing to secondary criticality problems, it becomes difficult to envision an autocatalytic sequence in a molten plutonium system.

The fact that the inherent safety of these systems is assured through effects much larger than the Doppler effect is a safety advantage and a design convenience, considering the present difficulty in obtaining reliable Doppler effect estimates.[(33)]

Future Developments

Further developments which may be explored to investigate increased potential of this fuel system include: venting of capsules to relieve pressure limitation and to reduce in-core fission product inventories; direct contact of coolant and fuel to relieve thermal limitations and to capitalize on fuel mobility; and lower neutron cross-section containment to increase breeding capability. Another major area which must be considered in order to utilize the potential of these systems is blanket development. The molten plutonium systems presently being considered are strictly external breeders, so that the blanket truly deserves equal prominence with the core. Far less work, however, has been done in this area.

Summary of the Molten Plutonium Programme

A core fuelled with molten plutonium (LAMPRE I) has been operated successfully for over 2 years. A versatile ternary fuel has been selected which appears promising for use in a fast breeder core. The reliable containment of this fuel appears to be feasible. In the present concepts, a tantalum container is used. The associated parasitic neutron losses are tolerable in a hard spectrum, if the plutonium content of the fuel is properly selected as a compromise between heat transfer and breeding potential.

The major potential advantages for molten plutonium fuelled reactors lie in safety, fuel element lifetime, and performance.

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